

UNIVERSITY OF VICTORIA, BC, Canada

WMO: 717830

Lat: 48.4569N

Lon: 123.3047W

Elev: 60

StdP: 100.61

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.4	-0.5	-11.1	1.5	-0.5	-7.5	2.0	1.7	7.1	4.4	6.2	4.6	2.6	30	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	27.0	18.0	25.2	17.4	23.5	16.7	19.1	25.0	18.2	23.4	17.3	22.2	2.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.2	20.9	16.1	11.5	19.9	15.3	11.0	19.1	54.5	25.1	51.4	23.6	48.9	22.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.7	4.2	DB	-5.1	30.8	2.3	2.0	-6.7	32.2	-8.1	33.4	-9.4	34.5	-11.1	35.9	(4)
(5)			WB	-6.0	20.4	2.6	1.3	-7.9	21.3	-9.5	22.1	-11.0	22.8	-12.9	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	5.9	5.9	7.4	9.6	12.6	15.2	17.5	17.4	14.8	10.6	7.6	5.5	(6)
(7)		DBStd	5.00	2.82	2.59	2.38	2.27	2.50	2.27	2.17	1.97	2.20	2.15	2.75	2.86	(7)
(8)		HDD10.0	607	129	115	86	33	5	0	0	0	18	79	142	(8)	
(9)		HDD18.3	2762	387	348	339	262	177	99	42	41	107	238	323	398	(9)
(10)		CDD10.0	923	1	1	4	21	87	155	233	231	146	37	7	1	(10)
(11)		CDD18.3	38	0	0	0	0	1	4	17	14	2	0	0	0	(11)
(12)		CDH23.3	403	0	0	0	0	9	57	178	138	21	0	0	0	(12)
(13)		CDH26.7	71	0	0	0	0	1	11	37	20	2	0	0	0	(13)
(14)	Wind	WSAvg	1.8	2.0	2.0	2.0	2.0	1.8	1.8	1.6	1.5	1.4	1.5	1.8	1.9	(14)
(15)	Precipitation	PrecAvg	646	107	73	48	34	26	19	14	21	33	61	98	112	(15)
(16)		PrecMax	873	217	171	109	72	52	68	38	67	72	207	191	264	(16)
(17)		PrecMin	389	16	11	12	17	4	2	0	0	4	4	16	10	(17)
(18)		PrecStd	125	53	39	27	14	12	15	11	17	18	46	44	54	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	12.4	15.9	20.4	24.7	28.4	30.3	29.0	25.8	19.1	14.5	13.2	(19)
(20)			MCWB	11.2	9.5	11.0	12.9	15.5	17.9	19.9	18.3	17.0	14.3	12.8	12.4	(20)
(21)		2%	DB	11.2	11.1	13.4	17.2	21.9	24.8	27.2	26.6	23.3	16.9	13.0	11.3	(21)
(22)			MCWB	10.2	8.8	10.0	12.0	14.6	16.8	18.4	18.0	16.3	13.1	11.8	10.2	(22)
(23)		5%	DB	10.2	10.1	12.1	15.2	19.7	22.3	25.2	24.7	21.4	15.3	12.0	10.1	(23)
(24)			MCWB	9.4	8.4	9.2	10.8	13.9	15.9	17.5	17.6	15.8	12.5	11.0	9.1	(24)
(25)		10%	DB	9.4	9.2	11.0	13.7	17.8	20.2	23.4	22.9	19.5	14.0	11.2	9.1	(25)
(26)			MCWB	8.6	7.9	8.7	10.0	13.0	14.7	16.9	16.9	15.2	11.9	10.3	8.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	10.7	12.1	14.2	16.7	18.9	20.9	20.1	18.5	16.6	13.6	12.8	(27)
(28)			MCDWB	12.2	11.2	13.9	18.2	21.3	26.3	27.7	25.8	23.4	17.5	14.0	13.0	(28)
(29)		2%	WB	10.6	9.6	10.8	12.7	15.4	17.4	19.4	19.0	17.5	14.3	12.2	10.7	(29)
(30)			MCDWB	10.9	10.3	12.3	16.2	20.1	23.4	25.8	24.4	21.5	15.5	12.6	11.0	(30)
(31)		5%	WB	9.7	8.9	9.9	11.6	14.4	16.3	18.3	18.1	16.6	13.2	11.2	9.4	(31)
(32)			MCDWB	10.0	9.6	11.4	14.4	18.7	21.3	23.8	23.3	20.0	14.3	11.7	9.9	(32)
(33)		10%	WB	8.8	8.2	9.1	10.7	13.4	15.3	17.2	17.3	15.6	12.2	10.4	8.4	(33)
(34)			MCDWB	9.2	8.9	10.5	13.0	17.1	19.5	22.3	22.0	18.5	13.5	11.0	8.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	5.0	6.2	7.5	9.0	9.5	11.0	10.7	8.9	6.3	4.8	4.2	(35)
(36)			MCDBR	4.5	5.5	7.3	10.0	12.6	13.5	14.7	14.1	12.0	8.0	5.0	5.0	(36)
(37)			MCWBR	3.7	4.0	4.6	5.6	6.5	6.4	6.8	6.5	6.0	5.0	4.0	4.3	(37)
(38)		5% WB	MCDBR	4.3	4.8	6.6	9.0	11.2	12.3	13.4	12.7	10.4	6.7	4.8	4.8	(38)
(39)			MCWBR	3.8	3.8	4.5	5.5	6.2	6.3	6.7	6.5	5.9	5.0	4.2	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.317	0.331	0.358	0.377	0.355	0.348	0.353	0.339	0.331	0.325	0.314	(40)	
(41)		taud	2.526	2.550	2.525	2.441	2.402	2.482	2.539	2.522	2.561	2.565	2.507	2.490	(41)	
(42)		Ebn at Noon	759	850	891	894	884	906	908	889	872	820	738	710	(42)	
(43)		Edh at Noon	61	73	87	104	113	105	98	95	83	71	61	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.81	2.83	4.29	5.49	6.00	6.55	5.53	3.94	2.17	1.11	0.75	(44)	
(45)		RadStd	0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.32	0.24	0.17	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	-0.61	-1.12	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page